COVID-19 Early Case Trend Analysis & Recovery Insights

1. Introduction

This project is based on a case study provided by HealthGuard Analytics Pvt. Ltd., a healthcare data analytics firm that delivers insights to government health departments and hospitals.

The objective of this study is to analyze early-stage COVID-19 patient data to understand infection patterns, demographic characteristics, recovery behavior, and regional trends. These insights are intended to support data-driven public health decisions.

2. Business Context

HealthGuard Analytics collaborated with a national public health authority to analyze patient-level data from the early phase of a COVID-19 outbreak.

The dataset includes confirmed cases along with demographic details, infection history, exposure metrics, and patient outcomes such as released, isolated, or deceased.

The analysis aims to improve:

- Screening strategies
- Isolation planning
- Early outbreak response systems

3. Problem Statement

The public health authority seeks answers to the following key questions:

1. Who is getting infected?
   Analysis of demographic patterns including age, gender, country, and region.
2. How are infections spreading?
   Identification of infection reasons, infection order, and exposure levels.

3. What are the recovery trends?
   Evaluation of recovery duration from confirmation to release.
4. Which regions are most impacted?
   Comparison of confirmed and recovered cases across regions.
5. What factors influence recovery time?
   Investigation of whether age, contact number, or infection order affects recovery duration.

4. Analytical Objectives

The project focuses on the following analytical tasks:

- Perform Exploratory Data Analysis (EDA) to understand data structure and quality
- Apply descriptive statistics to summarize patient demographics and outcomes
- Visualize trends related to:
    - Gender distribution
    - Age distribution
    - Regional case concentration
    - Infection sources
    - Recovery timelines
- Calculate recovery duration using confirmed and released dates
- Apply Linear Regression to:
    - Predict recovery time
    - Identify influencing factors

5. Dataset Description

The dataset consists of patient-level records with the following attributes:

5.1 Demographics

- Sex
- Birth Year
- Country
- Region

5.2 Infection Details

- Infection Reason
- Infection Order
- Infected By

5.3 Exposure Metrics

- Contact Number

5.4 Timeline Data

- Confirmed Date
- Released Date
- Deceased Date

5.5 Case Outcome

- State (Released, Isolated, Deceased)

6. Methodology

6.1 Data Cleaning

- Converted date columns into datetime format
- Derived age from birth year
- Handled missing values by filtering relevant subsets
- Created separate datasets for EDA and recovery analysis

6.2 Feature Engineering

- Created recovery_days as the difference between released and confirmed dates
- Categorized patients into age groups
- Prepared datasets for machine learning

6.3 Exploratory Data Analysis

- Analyzed gender and age distribution
- Identified regional hotspots
- Studied infection sources and spread patterns

6.4 Recovery Analysis

- Evaluated recovery time distribution
- Compared recovery across age and gender
- Performed correlation analysis

6.5 Machine Learning

- Applied Linear Regression
- Applied Random Forest Regressor
- Compared model performance using R^2 score and Mean Absolute Error

7. Results and Findings

7.1 Demographic Analysis

- The most affected age group was 36–60 years
- Gender distribution was relatively balanced

7.2 Regional Analysis

- The highest number of cases were observed in:
    - Gyeongsangbuk-do

    - Capital Area

7.3 Infection Spread Analysis

- Major infection sources included:
    - Contact with infected individuals
    - Travel-related exposure
- A significant portion of data was labeled as unknown due to missing values

7.4 Recovery Trends

- Average recovery time was approximately 15 days
- Most patients recovered within 7 to 24 days

8. Machine Learning Insights

- Random Forest performed better than Linear Regression
- Age showed a stronger influence on recovery time compared to gender
- Model performance was limited due to small dataset size

9. Limitations

- High number of missing values in infection-related attributes
- Very limited recovery dataset (approximately 28 samples)
- Results are exploratory and may not generalize

10. Expected Outcomes

The analysis provides:

- Understanding of early infection patterns
- Identification of high-risk demographic groups

- Insights into recovery timelines
- Data-driven support for public health decision-making

11. Conclusion

This project demonstrates the application of data analytics and machine learning in understanding infectious disease trends.

Despite limitations in data completeness, the analysis provides meaningful insights into infection spread, demographic impact, and recovery patterns. These findings can assist healthcare authorities in improving response strategies during early outbreak stages.

12. Author

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